

Machine Learning trains computers to learn from data without explicit programming.

Supervised learning uses labeled datasets to train algorithms for predictions.

Unsupervised learning finds hidden patterns in data without pre-existing labels.

Reinforcement learning rewards algorithms for making the right autonomous choices.

Data is the most critical foundation for training accurate models.

Overfitting occurs when a model performs poorly on new data.

Underfitting happens when a model is too simple to learn.

Deep Learning is a subset of machine learning using neural networks.

Neural networks mimic the structure of the human brain's neurons.

Features are individual independent variables acting as inputs for models.

Labels represent the final output or target variable we predict.

Regression models predict continuous numerical values like house market prices.

Classification models categorize data into distinct groups like spam email.

Clustering groups similar data points together without using prior labels.

Training data builds the model while test data evaluates performance.

Bias represents the simplifying assumptions a model makes to learn.

Variance shows how much the model output changes with data.

Algorithms require clean data to avoid generating flawed, inaccurate outputs.

Feature engineering selects and transforms raw variables into useful inputs.

Hyperparameters are external settings adjusted to optimize model training speed.

Natural Language Processing allows machines to understand human spoken language.

Computer Vision enables software to analyze and interpret digital images.

Decision Trees split data into branches to make logical choices.

Random Forests combine multiple decision trees to improve overall accuracy.

Gradient Descent is an optimization algorithm used to minimize errors.

Cross-validation splits data repeatedly to ensure stable model performance evaluation.

Anomalies and outliers can heavily distort machine learning model predictions.

Linear Regression establishes a straight-line relationship between two variables.

Machine learning models require regular updates to prevent accuracy drift.